

The empirical recurrence for OEIS A232373, recovered from its tail

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Abstract

OEIS A232373 records “Empirical recurrence of order 64” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 348$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the whole sequence, so the recurrence holds from $n = 65$ — every index at which it can be written down. Its last coefficient is $c_{64} = 370835456 \neq 0$, so no further tail satisfies anything shorter; any order-64 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A232373 is “Number of $n \times 5$ 0..3 arrays with every 0 next to a 1, every 1 next to a 2 and every 2 next to a 3 horizontally, diagonally or antidiagonally, and no adjacent values equal.”. It has offset 1 and begins

14, 1130, 32722, 927706, 27133338, 789866870, 22946925502, 667514680522, ...

The entry states:

Empirical recurrence of order 64 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 07:22:07, revision 7) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 348$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 348$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Here the stated order is already the minimal order of the sequence itself, so no tail has to be taken.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 696$ exact terms from $n = 1$ on, it returns order 64 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{64} c_i a(n-i),$$

c_1	18	c_{17}	312034124105	c_{33}	−2809780385853900	c_{49}	−215456328114250
c_2	189	c_{18}	1548459259881	c_{34}	3022198604295029	c_{50}	339491947580328
c_3	3091	c_{19}	3854642114585	c_{35}	−329200002730017	c_{51}	133585199067504
c_4	15059	c_{20}	2649502746303	c_{36}	1938530338335470	c_{52}	−206274795731604
c_5	174772	c_{21}	2315245738578	c_{37}	−2756611810370163	c_{53}	−53354718122056
c_6	1260827	c_{22}	−360554678662	c_{38}	1659296107499844	c_{54}	17132123029280
c_7	6809799	c_{23}	22985989053971	c_{39}	−3779135681749066	c_{55}	8801375240816
c_8	23697816	c_{24}	21324751563865	c_{40}	1383096701474737	c_{56}	30584731414400
c_9	37012260	c_{25}	−91598320613646	c_{41}	176815007055836	c_{57}	1660614089856
c_{10}	117681885	c_{26}	−167576990226071	c_{42}	3807405545707050	c_{58}	4330639809536
c_{11}	125290054	c_{27}	−433607901370072	c_{43}	−1190999340475876	c_{59}	1530478432768
c_{12}	48204634	c_{28}	517038142130103	c_{44}	−14482914380434	c_{60}	130789922560
c_{13}	−8275103872	c_{29}	199798813896997	c_{45}	−869376164915513	c_{61}	−230881627136
c_{14}	−36043953793	c_{30}	1566230738942686	c_{46}	−1171476820450679	c_{62}	−29579572224
c_{15}	−108449039378	c_{31}	−1599889666370854	c_{47}	−876824435986463	c_{63}	4612362240
c_{16}	−130787232621	c_{32}	564562135063869	c_{48}	−384488980769306	c_{64}	370835456

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 65$, and it is the recurrence OEIS A232373 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 1$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{64} - c_1 x^{64-1} - \dots - c_{64}$. Its constant term is $-c_{64} = -370835456$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 64 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 64, so $\deg q = 64$, and it is monic. It holds from some index t on. From $\max(t, 1)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 64, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 15 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 64, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 370835456.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-64 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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